radeon-fix

PRIVATE AI CODE-REPAIR AGENT ON AMD RADEON / ROCM

Track 2 | AMD AI DevMaster Hackathon 2026
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One local model. Your code never leaves the machine.

Issue-driven repair for private repositories: paste a bug report, get an auditable patch.

CLI

```
$ radeon-fix --repo <path>
             --issue "..."
```

Output:

`patch.diff  report.md  trajectory.json`
`run.json  gpu.json (rocm-smi)`

Exit: 0 verified | 1 no patch | 2 fatal | 3 rejected/failed check

ARCHITECTURE

1. Issue text
radeon-fix --repo --issue

2. Temporary git clone
fresh workspace, repo untouched

3. Aider Repo Map C1R1
2048-token map + one refresh

4. Qwen3-14B BF16 on ROCm
local vLLM, loopback endpoint

5. J3 xgrammar actions
Edit V2 exact replace

6. V1R1 verifier
deterministic checks, <=1 repair

GPU PERFORMANCE ON gfx1100 (51.52 GB VRAM)

TTFT p50 104 ms
p95 137 ms; decode p50 25.7 tok/s (10-request probe)

Prefix cache 18x / 77x
TTFT 3976->219 ms at 4K; 43,568->562 ms at 16K

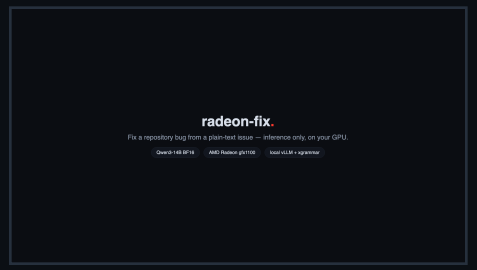
xgrammar 100% valid
189/189 structured actions schema-valid, zero JSON retries

Stable service
10/10 tool calls, 20/20 schema checks, 30/30 stability requests

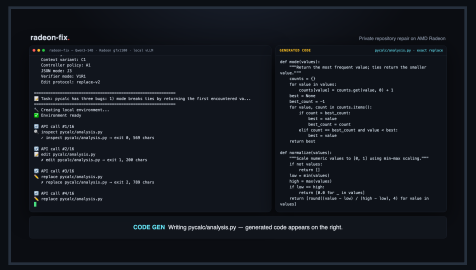
BF16 + INT8 KV
46.7 GB VRAM used, 32K context, no CPU offload

Agent-level: mean 1.92 min/task, P95 6.04 min
Dev gate: 5/5 patches, 5/5 applies, 38/38 valid actions

OUTPUT: patch.diff -> review -> git apply



1. Issue + workspace



2. Code generation in terminal



3. Patch + 7/7 tests pass